

Utkarsh Goel

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EDUCATION

New York University, Courant Institute of Mathematical Sciences

Sep 2025 – May 2027

Master of Science in Computer Science | GPA: 3.5 / 4.0 | Violet Internship & Research Award, Summer 2026

Coursework: Deep Learning, Machine Learning, AI Accelerators / Efficient AI, Honors Algorithms (PhD, Prof. Khot), Quantum Computing (PhD, Prof. Regev), Mathematical Tools for Neuroscience (PhD, Profs. Simoncelli and Landy), Operating Systems, Programming Languages

Manipal Institute of Technology, Manipal, India

Jul 2019 – May 2023

Bachelor of Technology in Computer and Communication Engineering | CGPA: 8.18 / 10

RESEARCH EXPERIENCE

Graduate Researcher, Regev Lab (Prof. Oded Regev), NYU Courant Institute

May 2026 – Present

- First author of "Folding It In: Structure-Aware Deep Splicing Models" (with A. Ramteke and O. Regev), accepted at NECB 2026.
 - Showed that adding predicted RNA secondary structure as extra input channels fixes a systematic blind spot in state-of-the-art splicing models (SpliceAI, Pangolin, AlphaGenome): higher held-out accuracy at the same parameter count, about half the structure-dependent error, and correct predictions on compensatory mutations that sequence-only models get wrong.
 - Built the full pipeline: RNAPfold structure features, multi-seed OpenSpliceAI retraining, and the MPRA-based bias analysis.
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EXPERIENCE

Software Development Engineer 1, Amazon (Alexa Automotive), Bangalore

Jan 2024 – Jul 2025

- One of 3 founding engineers on the migration of Alexa Automotive from a finite-state-machine dialog stack to a production LLM agent that reads live vehicle sensor data and calls vehicle APIs to complete multi-step requests.
- Shipped the agent from hackathon demo to production on AWS (Docker) after Audi, Rivian and Mahindra adopted it.
- Built and deployed Java/Spring Boot microservices (data registries, user management, IoT directives) serving millions of vehicles across Audi, Stellantis and Mahindra.

Product Analyst, Zoca (Accel-backed), Data Engineering Team, Bangalore

Jul 2023 – Dec 2023

- Single-handedly built the analytics stack for a sales-engagement product (Python data pipelines, Node.js backend); daily active users rose 50% month over month and SMB client revenue 20%.
- Trained and deployed scikit-learn lead-classification models that raised lead-generation accuracy 25%; automated QA with Python and Selenium, cutting manual test time 30%.

Machine Learning Engineer Intern, EliteFit.AI, Singapore

Aug 2022 – Jun 2023

- Built a computer-vision pipeline that normalizes user video across up to 25° of camera pan and zoom using optical flow and affine/homography transforms; frame detection reached F1 0.98 across 15+ exercise categories.
 - Built a body-part weighting scheme from open-source pose datasets to tune the cosine-similarity and cross-correlation metrics behind real-time form feedback for 1,000+ users.
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PROJECTS

Adaptive Block Allocation for Diffusion Language Models | Efficient AI, NYU Courant [\[GitHub\]](#)

- Trained a 1.1M-parameter MLP that reads a diffusion LM's hidden state at each block boundary and picks the next block size; on LLaDA-8B it beats the rule-based AdaBlock (ICLR 2026) by 2.8 pp mean accuracy on GSM8K, MATH and MBPP at matched per-token compute. Owned the predictor, training loop, heuristic labels and LLaDA evaluations (team of three).

LLM Pre-Training with RL Chain-of-Thought Reasoning | NYU Machine Learning Course [\[GitHub\]](#)

- Pre-trained a 100M-parameter transformer from scratch in PyTorch (custom self-attention, multi-GPU training) and fine-tuned it with reinforcement learning for chain-of-thought reasoning.

Robotics Perception Layer | AI4CE Lab, NYU [\[GitHub\]](#)

- Integrated five foundation models (CoTracker, FoundationPose, SAM2, Depth Anything, HaMeR) into one perception stack on the Apple EgoDex dataset, processing 30 fps RGB-D streams for dexterous-manipulation research.
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TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, Bash

Machine Learning: PyTorch, Transformers, LLM agents and tool calling, RL fine-tuning, computer vision, scikit-learn, multi-GPU training, quantization and pruning

Infrastructure: AWS (EC2, S3, Lambda), Docker, Spring Boot, Apache Airflow, Linux, Git